

4 Centre Of Gravity And Moment Of Inertia

4.1 Introduction:

Center Of Gravity And Centroid

The center of gravity of a body is the point at which the resultant gravitational force by the weights of all the particles of the body passes or the point at which the whole of its mass may be assumed to be concentrated. The centre of gravity of a plane figure which has area but no weight and where the whole area is assumed to be concentrated is known as the centroid of that area.

4.2 Location of Centroid

Analytical Method For The Location Of Centroid: Let us take a plane area as shown in

Fig. 4.1. Let $\bar{x}$ and $\bar{y}$ be the co-ordinates of centre of gravity of the area from OY and OX axis respectively. Divide the entire area A into small elementary areas $A_1, A_2, A_3, A_4 \dots$ etc. whose centroids are at distances of $x_1, x_2, x_3, x_4 \dots$ etc. respectively from the vertical axis OY. The algebraic sum of the moments of these elemental areas about Y axis = $A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4 + \dots$

The moment of the total area about y – axis = $A \bar{x}$

The algebraic sum of the moments of the element areas about any axis is equal to the moment of the total area 'A' about the same axis.

$$\therefore \frac{A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4 + \dots}{A_1 + A_2 + A_3 + A_4 + \dots} = \bar{x}$$

or

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4 + \dots}{A_1 + A_2 + A_3 + A_4 + \dots} = \frac{\sum Ax}{\sum A}$$

Similarly by equating moments of the elements of the elemental areas about x – axis with that of the total area we have

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3 + A_4 y_4 + \dots}{A_1 + A_2 + A_3 + A_4 + \dots} = \frac{\sum Ay}{\sum A}$$

Knowing $\bar{x}$ and $\bar{y}$, the position of centroid of the given figure is fixed.

4.3 Centroid of Regular Areas

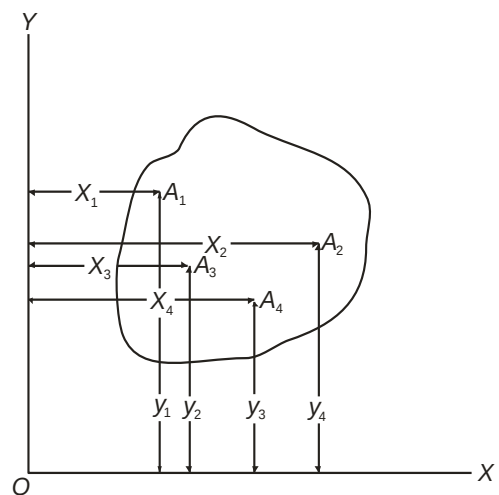


Fig. 4.1

(a) **Rectangle, square, parallelogram and rhombus:** These figures are having two axes of symmetry and their intersection point is the location of centroid.

(b) **Triangle:** the centroid is the point of intersection of medians. It is situated at $\frac{2}{3}$ of height from vertex or $\frac{1}{3}$ height from base.

(c) **Circle:** A circle is symmetrical about its diameters. The intersection of any two diameters, i.e. the centre of the circle is the centroid.

(d) **Semicircle:** A semicircle has one axis of symmetry. The centroid G lies at a distance of $\frac{4r}{3\pi}$ from its base, as shown in Fig. 4.2.

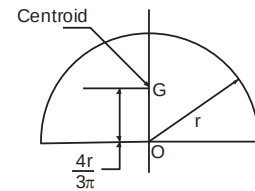


Fig. 4.2

4.4 Procedure for Locating the Centroid:

- (i) First observe whether the given area has any axis of symmetry or not.
- (ii) Divide the area into areas of regular shape whose centroids are known.
- (iii) Find out the areas of each broken up geometrical figure.
- (iv) Choose suitable axes of reference OX and OY (where there is one axis of symmetry, one of the axes chosen will coincide with the line of symmetry) with respect to which the position of centroid of the given figure is to be determined.
- (v) Find the distance of the centroids of the figure from the axes of reference.
- (vi) Take moments of the broken up areas about the axes of the reference.

First moment of area = area x perpendicular distance of centroid of respective area from the axes of reference.

$$= Ax \quad \dots\dots(5.1)$$

- (vii) The distance of the centroid from the axis of reference:

$$\bar{x} = \frac{\sum Ax}{A} \quad \text{and} \quad \bar{y} = \frac{\sum Ay}{A}$$

- (viii) $\bar{x}$ and $\bar{y}$ fix the position of the centroid of the given area.

4.5 Moment of Inertia

As per Newton's first law of motion the property of matter of resisting any change of its state of rest or motion in a straight line is known as inertia. For bodies in linear motion, the force required to change its state of motion shall depend upon the mass of the body. For bodies in

rotation about an axis, the force required to change the state of motion shall depend on the mass of the body and its distance from the axis around which it is rotating.

4.6 Moment of Inertia of an area

Moment of inertia of an area about x – x axis is written as I_{xx} and about y – y axis as I_{yy} . The moment of inertia of an area about an axis passing through its centroid is written by I_G or simply I . If the moment of area given by equation (5.1) is again multiplied by the perpendicular distance between C.G. of the area and axis OY i.e. distance x , then the quantity $(Ax).x = Ax^2$ is known as moment of moment of the area or second moment of area or moment of inertia about the axis OY. Similarly the moment of inertia about the axis OX = Ay^2 .

Now let us find out the moment of inertia of an irregular area A (Fig. 4.1) about x – x and y – y axis. Assume the given area to be composed of several elemental areas $A_1, A_2, A_3, A_4, \dots$ etc. and their distances from the two axes as $x_1, x_2, x_3, x_4, \dots$ etc. and $y_1, y_2, y_3, y_4, \dots$ etc. Then

$$I_{xx} = A_1 y_1^2 + A_2 y_2^2 + A_3 y_3^2 + A_4 y_4^2 + \dots = \sum Ay^2 \quad \dots(5.2)$$

$$I_{yy} = A_1 x_1^2 + A_2 x_2^2 + A_3 x_3^2 + A_4 x_4^2 + \dots = \sum Ax^2$$

.....(5.3)

4.7 Radius of Gyration

The radius of gyration of a body rotating about an axis is the distance from the axis where the whole area of the body is assumed to be concentrated so that the moment of inertia about the axis is unaltered. Now let us assume that the area shown in Fig. 4.3 is squeezed into a uniform strip of area A as shown in Fig. 4.4. Each elemental area will be then at the same distance k from the axis xx . The moment of inertia of this area is

$$I_{xx} = Ak^2. \quad \dots(5.3)$$

If the value of k is such that the moment of inertia given by Equation (5.2) remains the same as given by equation (5.4) then k is known as the radius of gyration of the section about the axis xx .

$$I_{xx} = Ak^2 \text{ or } k = \sqrt{\frac{I_{xx}}{A}} \quad \dots(5.4)$$

4.8 Theorems Involving Moment of Inertia of Plane Figures

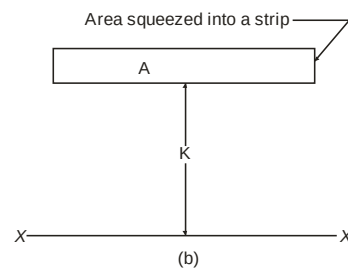


Fig. 4.3

4.8.1 Theorem of Parallel Axis

It states that “The moment of inertia of a figure about any axis in the plane of the figure is equal to the sum of the moment of inertia of that figure about the centroidal axis parallel to the given axis and the product of the area of the figure and square of the perpendicular distance between the two axes.”

Suppose the moment of inertia of a figure of area A as in Fig. 4.4 is to be determined about axis AB . Let G be the centroid of the figure. Draw a

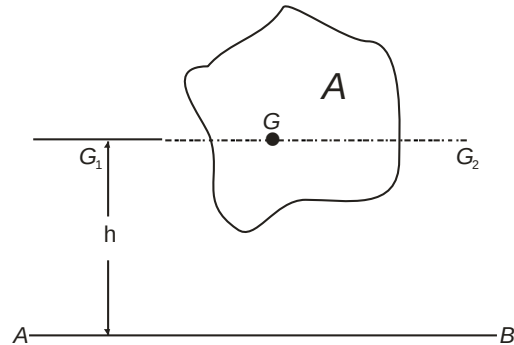


Fig. 4.4

centroidal line G_1G_2 parallel to AB . Then the moment of inertia of the figure about AB

$$I_{AB} = I_G + Ah^2 \quad \text{.....(5.5)}$$

where h is the distance between the two axes and I_G is moment of inertia of the area about axis G_1G_2 .

4.8.2 Theorem of Mutually Perpendicular Axis

It states that, “The moment of inertia of a figure about an axis perpendicular to the figure passing through its centroid is equal to the sum of its moment of inertia about two mutually perpendicular centroidal axes lying in the plane of the figure”.

Consider a plane figure of area A . Let two mutually perpendicular axes OX and OY be in the plane of the figure and passing through the centroid O as shown in Fig. 4.5.

Then the moment of inertia of the area about an axis OZ passing through the centroid O and perpendicular to the plane of the figure will be the sum of the moments of inertia of the area about the x -axis and y -axis, i.e.

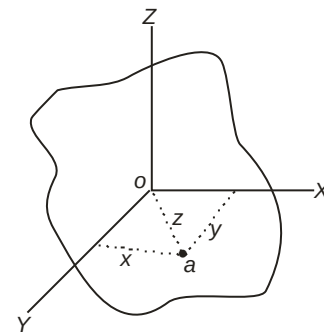


Fig. 4.5

$$I_{zz} = I_{xx} + I_{yy} \quad \text{.....(5.6)}$$

All the three axes OX , OY and OZ are mutually perpendicular to one another.

4.9 Moment of Inertia of Regular Sections

4.9.1 Rectangle

Consider a rectangle of breadth B and depth D as shown in Fig 4.6. Let the axes X-X and Y-Y pass through the C.G. Consider an elementary strip of width B and thickness dy at a distance y from X-X axis and parallel to it.

Area of the strip = B × dy

Moment of inertia of this elementary strip about X-X axis = B × dy × y²

∴ Moment of inertia of the whole rectangle about X-X axis (centroidal)

$$I_{xx} = \int_{-\frac{D}{2}}^{\frac{D}{2}} B \times dy \times y^2$$

$$I_{xx} = B \left[\frac{y^3}{3} \right]_{-\frac{D}{2}}^{\frac{D}{2}} = B \left[\frac{D^3}{24} + \frac{D^3}{24} \right] = \frac{BD^3}{12}$$

$$I_{xx} = \frac{DB^3}{12}$$

Similarly we can get $I_{yy} = \frac{BD^3}{12}$

Moment of inertia of the rectangle about its base AB will be obtained by applying the theorem of parallel axis.

$$I_{AB} = I_{xx} + B \times D \times \left(\frac{D}{2} \right)^2 = \frac{BD^3}{12} + \frac{BD^3}{4} = \frac{BD^3}{3}$$

4.9.2 Hollow Rectangle (Fig 4.7)

$$I_{xx} \text{ for the hollow rectangle} = \frac{BD^3}{12} - \frac{bd^3}{12}$$

Similarly

$$I_{yy} = \frac{DB^3}{12} - \frac{db^3}{12}$$

∴

$$I_{zz} = I_{xx} + I_{yy} = \frac{BD^3}{12} - \frac{bd^3}{12} + \frac{DB^3}{12} - \frac{db^3}{12}$$

$$= \frac{BD}{12} (B^2 + D^2) - \frac{bd}{12} (b^2 + d^2)$$

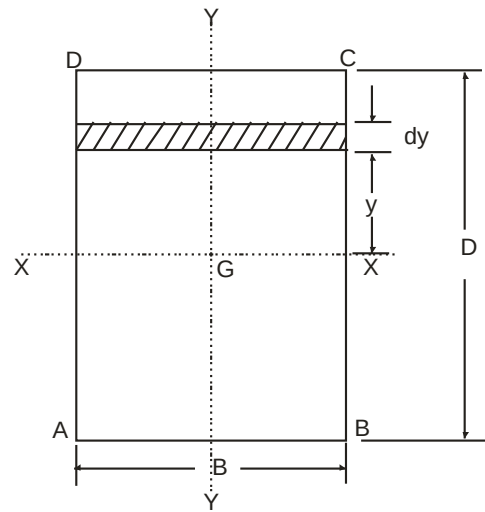


Fig. 4.6

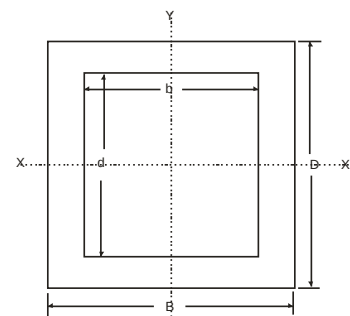


Fig. 4.7

4.9.3 Triangle

Consider a triangle ABC of base width B and height H. It is desired to find out the moment of inertia about the base BC and about the axis x-x passing through the centroid and parallel to the base as shown in the Fig. 4.8.

Consider an elementary strip of thickness dy at a distance y below the vertex A. Triangle ABC and ADE are similar.

$$\frac{DE}{BC} = \frac{y}{H}$$

∴

$$\text{or} \quad DE = \frac{BC \times y}{H} = \frac{by}{H}$$

$$\text{Area of the strip} = \frac{By}{H} \times dy$$

Moment of inertia of the strip about the base = area $\times$ (distance)²

$$= \frac{By}{H} \times y dy \times (H - y)^2 = \frac{B}{H} \cdot y (H^2 + y^2 - 2Hy) \cdot dy$$

$$= \frac{B}{H} (H^2 y + y^3 - 2Hy^2) dy$$

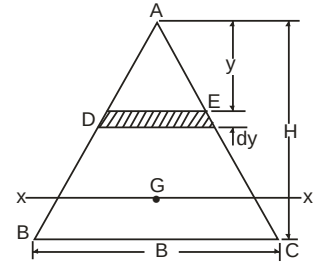


Fig. 4.8

Moment of inertia of the triangle about the base BC is given by

$$I_{BC} = \int_0^H \frac{B}{H} (H^2 y + y^3 - 2Hy^2) dy = \frac{B}{H} \left[\frac{H^2 y^2}{2} + \frac{y^4}{4} - \frac{2Hy^3}{3} \right]_0^H$$

$$= \frac{B}{H} \left[\frac{H^4}{2} + \frac{H^4}{4} - \frac{2H^4}{3} \right] = \frac{BH^3}{12}$$

Moment of inertia of the triangle about an axis x-x passing through the centroid and parallel to the base is found out as follows:

Now from the theorem of parallel axis we have

$$I_{BC} = I_G + \frac{BH}{2} \times \left(\frac{H}{3} \right)^2$$

$$I_G = I_{BC} - \frac{BH^3}{18} = \frac{BH^3}{12} - \frac{BH^3}{18} = \frac{BH^3}{36}$$

4.9.4 Circle

Consider a circle of radius R as shown in Fig 4.9. Consider an elementary ring of thickness dr at a radius r from o. Area of this ring = $2\pi r \cdot dr$.

Moment of inertia of this ring about z axis passing through o

and perpendicular to the plane of the given area = $2\pi \times r^2 dr$.

$\therefore$ Moment of inertia of the whole area

$$I_{zz} = \int_0^R 2\pi r^3 dr = 2\pi \int_0^R r^3 dr = 2\pi \left[\frac{r^4}{4} \right]_0^R = \frac{\pi R^4}{2}$$

$$\text{But} \quad I_{zz} = I_{xx} + I_{yy}$$

The given area is symmetrical about

xx as well as yy axis.

$$\therefore \quad I_{xx} = I_{yy}$$

$$I_{zz} = 2 I_{xx} = 2 I_{yy}$$

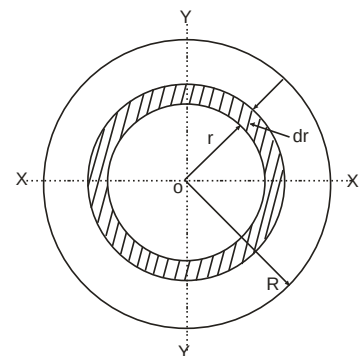


Fig. 4.9

$$\text{or } I_{xx} = I_{yy} = \frac{I_{zz}}{2} = \frac{\pi R^4}{4} = \frac{\pi D^4}{64}$$

For a hollow circular area of external diameter D and internal diameter d.

$$I_{xx} = I_{yy} = \frac{\pi D^4}{64} - \frac{\pi d^4}{64} = \frac{\pi}{64} (D^4 - d^4)$$

4.10 Section Modulus and Polar Moment of Inertia

It is defined as ratio of the moment of inertia of the section about the axis passing through the centroid and the distance of extreme fiber of the section from the centroidal axis. It is written as Z. Every section has got two section moduli.

For rectangle shown in Fig. 4.6

$$Z_{xx} = I_{xx} / \frac{D}{2}$$

$$\text{and } Z_{yy} = I_{yy} / \frac{B}{2}$$

Polar moment of inertia of a body is the moment of inertia of the body about its centroidal axis which is perpendicular to the plane of the body.